2018 Ph H2 Q10

Section: Particles and Waves

Topic: Radiation, The Inverse Square Law

A detector is used to measure the count rate from a radioactive point source.
The detector is moved further away from the source.

(a) The count rate is found to decrease.

Explain why this happens, using the inverse square law.

✅ As the detector moves farther from the source, the **radiation spreads out** over a larger area.
The **intensity** (and therefore the count rate) falls with the **square of the distance**:

$$I \propto \frac{1}{r^2}$$

So if the distance is **doubled**, the count rate becomes **one quarter** of its original value.

(b) At a distance of 0.20 m from the source, the count rate is 640 counts per second.

Calculate the expected count rate at a distance of 0.50 m from the source.

Use the inverse square law:

$$\frac{I_1}{I_2} = \left(\frac{r_2}{r_1}\right)^2$$

$$\frac{640}{I_2} = \left(\frac{0.50}{0.20}\right)^2 = (2.5)^2 = 6.25 \Rightarrow I_2 = \frac{640}{6.25} = 102 \text{ counts/s}$$

🧠 Revision Tips

- Radiation intensity from a **point source** follows the **inverse square law**:
$$I \propto \frac{1}{r^2}$$
- Doubling distance = intensity becomes a **quarter**
- Use ratios to calculate new count rates:
$$\frac{I_1}{I_2} = \left(\frac{r_2}{r_1}\right)^2$$
- Count rate is a proxy for **intensity** if all other factors (e.g. efficiency) stay constant